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Genomic Landscape and Precision Therapy in Prostate Cancer: Current Status and Future Directions

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Abstract

Prostate cancer continues to impose a significant health burden worldwide and in the United States, with rising incidence, persistent mortality rates, and disparities in incidence rates and outcomes. Advances in genomic profiling have highlighted the remarkable heterogeneity of prostate cancer, revealing frequent alterations in key oncogenes, tumor suppressors, and pathways such as androgen receptor signaling and DNA damage repair that drive disease progression and therapy resistance. These molecular discoveries and their mechanistic validation in laboratory and preclinical models are transforming prostate cancer management, enabling more precise, biomarker-driven treatments and clinical trials. However, challenges, including intra- and inter-tumor genomic diversity, remain formidable barriers to progress. This review provides an overview of the current genomic landscape of prostate cancer and discusses recent advances in precision therapy. We also discuss emerging directions in integrating molecular profiling with personalized patient care, while addressing the roadblocks hampering progress. The discussion highlights the critical need for continued innovation in the development of personalized therapies to improve the clinical outcomes for patients with prostate cancer.

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Keywords: Prostate cancer, Genetic alterations, Epigenetic modifications, Precision oncology, Neuroendocrine differentiation, Lineage plasticity, Tumor mutational burden, Microsatellite instability, Germline variants, Hereditary Prostate Cancer

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Introduction

Prostate cancer remains a leading cause of cancer-related morbidity and mortality worldwide. In the United States alone, it accounts for nearly 30% of all new cancer cases in men each year, making it the most frequently diagnosed non-cutaneous cancer [1]. According to the American Cancer Society, an estimated 313,780 new prostate cancer cases are projected to be diagnosed in 2025, with approximately 35,770 men expected to succumb to the disease [2]. Recent data indicate a concerning reversal in the decline of prostate cancer incidence rates observed in earlier years (2007-2014), with an average increase of about 3.0% each year observed between 2014 and 2021, particularly in advanced-stage diagnoses [1]. Moreover, mortality rates have not seen a substantial decline over the past decade despite improvements in early detection and treatment, emphasizing the critical need for continued advances in prevention, diagnosis, and precision therapy. Additionally, racial disparities remain pronounced, with Black men experiencing nearly twice the mortality rate compared to White men, highlighting the need for focused research to find ways to reduce the disparity gaps.

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The management of prostate cancer has remained challenging because of its significant genomic heterogeneity, influencing disease progression and treatment response, and often resulting in disease relapses. Advances in high-throughput sequencing and comprehensive genomic profiling have revealed numerous clinically actionable somatic mutations in advanced and metastatic prostate cancer. These include mutations in critical pathways such as androgen receptor (AR) signaling, DNA damage repair (DDR), PI3K/AKT signaling, and in key tumor suppressors, such as TP53 and RB1 [3]. These molecular insights have opened new avenues for therapeutic development and more effective clinical management by guiding individualized treatment

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strategies and the design of precision-focused clinical trials. Additionally, recent insights into tumor mutational burden and associated mechanisms are beginning to inform selection for immunotherapeutics in castration-resistant disease settings. However, inter- and intra-patient genomic diversity continue to pose challenges to the development of effective precision treatments.

This review summarizes the current status of the genomic landscape in prostate cancer and discusses advances in precision therapy. We also discuss emerging directions in integrating molecular profiling with personalized patient care, highlighting opportunities and obstacles for the next generation of genomic and precision oncology in prostate cancer.

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Genomic and epigenomic alterations in primary, metastatic, and castration-resistant prostate cancer.

Prostate cancer is a progressive disease that often develops from precancerous high-grade prostatic epithelial neoplasia (HGPIN) to invasive and potentially metastatic prostate carcinoma. Metastatic prostate cancer is treated with androgen-deprivation therapy (ADT), either medical or surgical castration, coupled with an oral androgen receptor pathway inhibitor as the first line of care in most clinical situations. While most patients respond initially to this therapy, eventual progression to castration-resistant prostate cancer (CRPC) is inevitable [4]. During this evolutionary journey, prostate cancer cells acquire a diverse array of genetic and molecular alterations, including gene fusions, point mutations, deletions, amplifications, as well as transcriptomic and epigenetic changes (**Table 1**), many of which correlate with disease stage, therapeutic resistance, and clinical outcomes (**Fig. 1**).

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Gene fusions:

Gene fusions or rearrangements are frequently reported in prostate cancer and are generally considered early events in prostate carcinogenesis. The most common fusions involve androgen-responsive genes and ETS transcription factors, such as ERG, which help tumor cells acquire a less differentiated, more aggressive phenotype. The TMPRSS2-ERG fusion is present in approximately 50% of prostate cancer cases [5] and is shown to promote tumor aggressiveness by increasing vascular density, upregulating hyaluronan and platelet-derived growth factor-beta (PDGF-β), and downregulating caveolin-1 [6]. It also leads to the induction of transcription factors, HOXB13 and FOX1A1, which can further participate in prostate carcinogenesis and aggressive disease progression [5]. TMPRSS2-ERG engages in a positive feedback loop by inducing ERR-alpha, which subsequently upregulates ERG to support tumor proliferation [7]. It also enhances the metastatic progression through the upregulation of Metalloproteinase 9 (MMP9) and Plexin A2 (PLXNA2) [8]. In addition, TMPRSS2-ERG likely contributes to castration-resistance by facilitating AR signaling through maintenance of coregulator complexes at AR-bound sites across the genome [9]. Clearly, these data identify TMPRSS2-ERG as a common driver of metastatic and castration-resistant prostate cancer. In other reports, TMPRSS2 is shown to fuse with other ETS family transcription factors such as ETV1, ETV4, and ETV5, although it is less common than TMPRSS2-ERG [10-12]. These fusions occur in a smaller subset of prostate cancers and are often mutually exclusive with TMPRSS2-ERG, indicating alternative oncogenic routes within ETS-driven prostate tumorigenesis.

Gene mutations:

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In addition to the gene fusion, genetic mutations are also frequently reported in prostate cancer. A sequencing study by Barbieri et al. revealed mutations in *SPOP*, *FOXA1*, and *MED12* in prostate cancer, with *SPOP* mutations being the most common [13]. Another study by Hernandez-Llodra et al. detected SPOP protein loss in 42.9% of prostate cancer cases, with a prevalence of 60% in Grade 5 cases. Further, in combination with PTEN loss, it predisposes patients to the development of high-grade tumors as well as perineural invasion [14]. SPOP is an adapter of the CULLIN3/RING-box1 E3 ubiquitin ligase complex, mediating the polyubiquitination and degradation of various oncogenic proteins, including AR and c-MYC, that play a significant role in prostate cancer progression [15]. *SPOP* mutations also activate the PI3K/mTOR pathway, further promoting tumorigenesis [16]. Interestingly, SPOP mutations are almost exclusively found in ETS fusion-negative prostate tumors [13]. The mutual exclusivity of ETS rearrangements and SPOP mutations is a significant observation that suggests they serve as alternative oncogenic drivers, and that activating one pathway relieves selective pressure on the other.

FOXA1, a regulator of chromatin remodeling, exhibits point mutations associated with SPOP. FOXA1 acts as a pioneer transcription factor, facilitating AR binding in regions with condensed chromatin and thus defining AR transcriptional programs [17]. Therefore, FOXA1 mutations may alter the AR cistrome by modulating its genomic specificity and, therefore, affecting tumor phenotypes. A study by Tam *et al.* showed that *FOXA1* mutations were present in 35% of CRPC, where FOXA1 acted as an upstream regulator of semaphorin 3C (SEMA3C), causing its overexpression [18]. SEMA3C acts as a growth factor, driving growth and development of treatment resistance in prostate cancer [19]. Hernandez-Llodra et al. reported a correlation between *SPOP* and *FOXA1*, where both were associated with faster biochemical recurrence in ERG^{wt}

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prostate cancer following treatment [20]. In another study, *FOXA1* mutations were shown to cause dysregulation of the WNT pathway, suggesting that FOXA1 can have a multifocal effect on tumorigenesis [21].

A meta-analysis by Maddah et al. demonstrated that TP53 mutations were associated with the worst clinical outcomes, including a 13% decrease in one-year overall survival and a 20% and 16% decrease in three- and five-year overall survival, respectively [22]. *RB1*, which plays a role in cell cycle regulation, is also either mutated or lost in prostate cancer and linked to the development of advanced prostatic adenocarcinoma and neuroendocrine prostate cancer (NEPC) [23]. In preclinical mouse models, *RB1* loss was associated with increased metastatic potential, while *TP53* mutation was significantly associated with resistance to antiandrogen therapy through the upregulation of AR reprogramming factors, SOX2, and EZH2 [24]. Thus, alterations of *TP53* and *RB1* contribute to treatment resistance, metastasis, and phenotype evolution of advanced prostate cancer.

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Gene deletions:

One of the most common gene deletions reported in prostate cancer is *PTEN*, located on chromosome 10q (10q23.3 deletion). It is a negative regulator of the PI3K/Akt/mTOR signaling pathway that supports cell cycle progression and survival. Loss of PTEN is also associated with resistance to androgen receptor blockade [25, 26]. PI3K engages in crosstalk with AR signaling, and the upregulation of PI3K/Akt/mTOR is also linked to a reduced tumor immune response, including the blockade of type I interferons (IFN-α and IFN-β) and immune anergy [25, 26]. Latini et al. showed that differences in observed loss of heterozygosity at the *PTEN* region of 10q23.3

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were not significant between a control group, benign prostatic hyperplasia, and adenocarcinoma; however, a significant difference in loss of heterozygosity was observed in late-stage metastatic prostate carcinoma, suggesting its greater role in later stages of prostate carcinogenesis [27]. A study by Suzuki et al. demonstrated *PTEN*/*MMAC1* alterations (deletions or point mutations) in at least 1 metastatic site in 12 of the 19 patients studied, with 2 cases having homozygous deletions and 4 harboring point mutations, of which one mutation was common in primary and metastatic lesions and the remaining were detected sporadically [28]. These findings suggest *PTEN*/*MMAC1* alterations are frequent with substantial mutational heterogeneity among different metastatic sites, even within the same patient.

Another gene commonly deleted in prostate cancer is NKX3.1, located in the 8p21.2 region [29]. NKX3.1 encodes a homeobox transcription factor that regulates prostate epithelial cell differentiation and DNA repair. NKX3.1 promotes apoptosis by upregulating TNF-alpha and caspase-3 [30]. It also suppresses the fusion of TMPRSS2 with ERG, thus its loss is associated with an increased TMPRSS2-ERG fusion risk [31]. Sooreshjani et al. showed that LIMK2 acted as a negative regulator of NKX3.1, and NKX3.1 downregulation led to LIMK2 polyubiquitination, forming a negative feedback loop. The negative crosstalk-induced signaling between AR, its splice variant form (AR-V7), and AKT promotes aggressive phenotypes and castration resistance [32].

CHD1, encoding chromodomain-helicase DNA-binding protein 1, is among the more frequently altered chromatin regulators in prostate cancer and is commonly deleted through focal loss at the 5q21 locus in a significant subset of tumors [33]. It plays a role in chromatin remodeling and helps maintain open chromatin architecture, thereby enabling lineage-appropriate transcriptional

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programs. Loss of CHD1 results in widespread transcriptional reprogramming and altered AR-driven gene expression, underscoring its role in shaping functional molecular states [34]. Beyond transcriptional regulation, CHD1 also plays a critical role in the DNA damage response by facilitating recruitment of homologous recombination (HR) repair machinery to sites of double-strand breaks. Consequently, CHD1 loss impairs HR, promotes genomic instability, and induces a BRCA-like repair defect [35]. Thus, CHD1-deficient tumors may exhibit enhanced sensitivity to PARP inhibitors and other DNA-damaging agents, as also suggested in preclinical studies [36, 37]. CHD1-deleted tumors are typically ETS fusion-negative and frequently co-occur with SPOP mutations, suggesting coordinated disruption of chromatin regulation and ubiquitin-mediated proteostasis [14, 38, 39]. This genomic context highlights how CHD1 loss may contribute to tumor evolution by establishing a permissive chromatin landscape that facilitates transcriptional plasticity.

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Gene amplifications:

Gene amplifications are a common feature in advanced prostate cancer and are strongly associated with disease progression, treatment resistance, and poor prognosis. Among these, MYC, AR, and MYB amplifications have emerged as key contributors to aggressive tumor phenotypes and castration resistance. MYC amplification is observed in prostate cancer, particularly in advanced and metastatic cases, playing a critical role in tumor progression and aggressiveness [40]. It is shown that MYC overexpression can disrupt the AR signaling, fostering a more aggressive, castration-resistant phenotype [41]. Furthermore, MYC amplification has been associated with immune evasion, suggesting that patients harboring such alterations might benefit from immune-modulatory therapies [42]. While MYC amplification is detectable in some high-grade prostatic

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intraepithelial neoplasia (HGPIN) lesions [43], Fromont et al. reported MYC overexpression in 29% of prostate cancer cases, exhibiting an association with a greater risk of metastasis [42]. The study also found a positive correlation between MYC overexpression and high Gleason score, with a <3% association with Gleason score 6 but a 66% association in Gleason score ≥8. Furthermore, Jin et al. found that a higher frequency of MYC overexpression in patients of African descent compared to those of European descent (12.9% compared to 10.2%), suggesting its role in racial disparities in prostate cancer outcomes [44].

AR amplification is also observed in prostate cancer, particularly in castration-resistant prostate cancer (CRPC), and contributes to the therapy resistance [45, 46]. AR amplification leads to higher AR protein levels, enabling cancer cells to maintain AR signaling even at very low androgen levels [46]. AR amplification is strongly associated with resistance to oral androgen receptor pathway inhibitors such as enzalutamide and abiraterone, contributing to treatment failure and disease progression [47]. Furthermore, co-amplification of genes in the vicinity, such as OPHN1, may also be significant in the development of CRPC.

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MYB is a transcription factor and a regulator of stemness, cellular differentiation, and survival [48, 49]. MYB amplification was reported in 26% of prostate cancer cases, with an enhanced frequency (53%) reported in the hormone-refractory disease [50]. MYB plays a role in driving prostate cancer aggressiveness and castration resistance [51, 52]. MYB-overexpressing cells retain nuclear AR localization even in an androgen-deprived environment and enhance the transcriptional activation of AR-responsive genes, such as KLK3/PSA, through a direct interaction with AR [52]. In addition, MYB overexpression is more prevalent in Black/African American patients than in

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White/European American patients and is associated with higher tumor grade. Importantly, MYB is shown to be a better predictor of early biochemical recurrence than PSA at diagnosis and gleason score, highlighting its potential as a biomarker for risk assessment and a therapeutic target [53]. Together, these findings establish MYB as a central transcriptional driver of prostate cancer progression, linking aberrant AR signaling with racial disparities and therapeutic resistance, and underscore its strong potential as both a prognostic biomarker and a targetable vulnerability in advanced and castration-resistant disease.

Pygo2 amplification has emerged as a functionally relevant alteration in prostate cancer progression [54]. As a chromatin effector of Wnt/β-catenin signaling, Pygo2 promotes transcriptional activation through recognition of H3K4 methylation and recruitment of co-activator complexes [55, 56]. In prostate cancer, Pygo2 amplification and overexpression have been linked to aggressive disease, enhanced lineage plasticity, and resistance to therapy, in part through epigenetic reprogramming of oncogenic and developmental gene networks [54, 57, 58]. Recent studies further implicate Pygo2 in shaping the tumor immune microenvironment, where it contributes to immune evasion by suppressing cytotoxic immune infiltration and altering antigen presentation pathways [59]. Together, these findings position Pygo2 as a network builder between epigenetic control, tumor progression, and immune suppression in advanced prostate cancer.

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Epigenomic alterations:

EZH2 is a histone methyltransferase that regulates gene expression through histone trimethylation. Melling et al. reported EZH2 positivity in 56.6% of prostate cancers, exhibiting associations with high Gleason grades and early PSA recurrence [60]. They also found a strong correlation of EZH2

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positivity with TMPRSS2-ERG fusion. Glutathione S-Transferase Pi 1 (GSTP1) is an antioxidant protein whose inactivation *via* hypermethylation can lead to tumorigenesis in prostate cancer. *GSTP1* hypermethylation is among the most common genetic alterations in prostate cancer. A study by Henrique and Jeronimo found *GSTP1* hypermethylation in >90% of prostate cancer cases [61]. Loss of GSTP1 leads to accumulation of oxidative DNA base damage and sustains the survival of prostate cancer cells, following exposure to protracted oxidative stress [62]. Kanwal et al. report that tumors with *GSTP1* hypermethylation express higher levels of 8-oxo-2'-deoxyguanosine, a marker of oxidative stress, when treated with oxidative stressors [63]. Therefore, *GSTP1* hypermethylation serves as an important marker to possibly target prostate cancer with oxidative stress-causing chemotherapies.

Epigenetic regulation of several significant genes by microRNAs has also been reported in prostate cancer [64]. One recent example is *MYB*, a gene amplified in a subset of prostate cancer that is regulated by both transcriptional and posttranscriptional mechanisms. *MYB* is androgen-responsive, but at high doses of androgen, its expression was repressed through a posttranscriptional mechanism involving microRNA. Specifically, high doses of androgens induced miR-150, which bound to the 3'UTR of *MYB*, causing its repression [65, 66]. This biphasic regulation of *MYB* was involved in mediating a biphasic effect of androgens on the growth of prostate cancer cells, highlighting the significance of targeting MYB either directly or indirectly by targeting its upstream or downstream regulators. Alterations in several other oncogenic and tumor suppressive miRNAs in prostate cancer have also been reported [67]. MicroRNAs can also reciprocally regulate gene expression, affecting prostate tumor growth and therapeutic response [68]. Specific miRNAs may be involved in different stages of prostate carcinogenesis and

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progression, such as EMT (miR-1, miR-29b, miR-200), angiogenesis (miR-21, miR146a), anoikis resistance (miR-132), and tumor-host interactions (miR-1, miR-15/16, miR-3a) [69]. Prostate-derived miRNAs can also be packaged into extracellular vesicles and have broader implications in prostate cancer pathobiology, therapeutic planning, as well as being used as prognostic biomarkers for prostate cancer [70].

Neuroendocrine differentiation and lineage plasticity.

Neuroendocrine (NE) differentiation in prostate cancer is a process in which prostatic adenocarcinoma cells acquire neuroendocrine-like features **(Fig. 2)**. It represents an adaptive, evolutionarily acquired state driven by specific genomic and epigenomic programs rather than a separate disease classification. Loss of AR in these tumors, predictably, renders them unresponsive to androgen-deprivation therapy (ADT). Genetic analysis by Rodarte et al. revealed that during neuroendocrine transformation, cell cycle regulators RB1 and TP53 are often lost, while transcription factors ASCL1 and NEUROD1 are upregulated [71]. Additional transcription factors, such as SOX2, FOXA2, ONECUT2, and BRN2, are also critical drivers that are reported to promote stemness, epithelial-mesenchymal transition (EMT), and NE markers such as chromogranin A (CHGA) and synaptophysin (SYP) in an AR-independent manner [72-75]. Among these, FOXA2 in particular has been shown to facilitate chromatin remodeling to promote lineage switching, while BRN2 is shown to cooperate with SOX2 to activate neuronal genes [75]. Epigenetic remodeling is central to lineage plasticity. Notably, EZH2 regulates epigenetic changes driving NE transdifferentiation by catalyzing H3K27me3 methylation, which silences tumor suppressor genes while promoting the expression of NE-associated transcription factors ASCXL1 and NEUROD1 [76]. Amplification and overexpression of MYCN and AURKA also contribute

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to NE differentiation, with concurrent amplification observed in 40% of NEPC cases, compared to 5% of prostate adenocarcinoma [76]. MYCN-overexpressing tumors have an enriched expression of both embryonic stem cell and EMT program gene sets, while AURKA can stabilize MYCN by dimerizing with it. Emerging evidence suggests that MYCN remodels chromatin states and cooperates with epigenetic regulators to suppress lineage-restricted differentiation while enabling adaptive cell state transitions [76-78]. Through these mechanisms, MYCN facilitates phenotypic flexibility and supports the emergence of neuroendocrine phenotypes in response to developmental cues or therapeutic pressure, underscoring its central role in neuroendocrine lineage specification and plasticity. Such genetic and epigenetic reprogramming highlight the complex mechanisms underlying NE differentiation and therapy resistance. Thus, targeting NEPC requires a multifaceted approach to inhibit factors that sustain the NE phenotype, thereby effectively treating these resilient tumors [79].

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Tumor Mutational Burden and Microsatellite Instability.

Tumor mutational burden (TMB) measures the number of mutant antigens produced by a tumor and reflects genomic instability. High TMB or the presence of a large number of mutant antigens allows a greater number of neoantigens to be presented on MHC molecules, potentially strengthening the immune responses. Therefore, it can be a good predictor of tumor susceptibility to treatment with targeted immunotherapy, such as PD-1/PD-L1 or CTLA-4 inhibitors. Microsatellite Instability-High (MSI-H) and deficient mismatch repair (dMMR) are closely related molecular features, both reflecting a defect in the DNA repair system that normally corrects errors occurring during DNA replication [80]. Familial microsatellite repair gene mutations, often referred to as Lynch Syndrome, consist of deleterious mutations in genes such as *MLH1* and

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MSH2. Screening for these mutations has been vital in therapeutic planning, as MSI-H/dMMR qualifies a patient for treatment with the PD-1 inhibitor pembrolizumab. While prostate cancer cells typically exhibit low TMB, finding the rare cells with high TMB and/or the closely related but MSI-H/dMMR is of potential clinical significance. A study by Klümper et al. found a 2.5% prevalence of MSI-H/dMMR mutation in prostate cancer patients [81], while another study by Lenis et al. found a similar 2.8% prevalence with 65% of those patients experiencing ≥50% decline in PSA (PSA50) in response to pembrolizumab, an immune checkpoint inhibitor. In contrast, the patients without MSI-H/dMMR tumors showed no response to the same therapy [82]. These findings highlight the need for genomic screening to identify patients with MSI-H/dMMR tumors that may benefit from immunotherapy.

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Germline Mutations and Hereditary Prostate Cancer.

Germline mutations are key predictors of hereditary cancer risk. Many of these mutations are involved in the DNA repair process, making them easy targets for therapeutics. Below, we discuss reports on BRCA1/2, HOXB13, ATM, and CHEK genes that have been studied in relation to prostate cancer risk **(Fig. 3)**.

BRCA1/2:

BRCA1 and *BRCA2* mutations affect homologous DNA recombination to repair double-stranded DNA breaks and are associated with an increased hereditary risk of prostate cancer [83]. Other homologous recombination proteins, such as ATM and CHK2, are also frequently mutated in cancer syndromes, contributing to similar DNA repair deficiencies and elevated prostate cancer risk [84, 85]. Like many inherited conditions, *BRCA1* and *BRCA2* mutations have a greater

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association with specific ethnicities, particularly Ashkenazi Jewish populations [86]. Studies also suggest that BRCA1 mutations may be more common in African American men with metastatic prostate cancer compared to men of European descent [87]. According to a prospective cohort study by Nyberg et al., patients with a *BRCA1* mutation had a 2.83 standardized incidence ratio for prostate cancer, while patients with a *BRCA2* mutation had a 4.45 standardized incidence ratio for prostate cancer [88]. Patients with a *BRCA2* mutation were also found to have an increased association with higher Gleason scores (>7) when compared to those with *BRCA1* mutations and the general population. While *BRCA1* mutations are not associated with the disease severity as *BRCA2* mutations, it is still associated with a higher rate of severe disease than prostate cancer patients without a germline mutation [89]. A meta-analysis by Oh et al. found a lack of significant difference in overall survival between patients with a *BRCA1* mutation and the general population, but a decreased overall survival in patients with a *BRCA2* mutation, compared to the general population, with a hazard ratio of 2.21 [90]. Additionally, they found that *BRCA2* mutations conferred a higher prostate cancer risk, with an odds ratio of 2.64, compared to *BRCA1*, with an odds ratio of 1.35. This suggests that a *BRCA2* mutation is a strong predictor of risk, disease severity, and poorer survival outcomes.

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HOXB13:

HOXB13 is a homeobox transcription factor that is critical in embryonic prostate development [91]. The G84E mutation at the 17q21-22 region is a germline variant associated with a hereditary predisposition to prostate cancer [91]. Ewing et al. report that *HOXB13* germline mutations are present in 1.4% of men with prostate cancer, compared with 0.1% of the general population, indicating a significant enrichment in affected individuals. *HOXB13* mutations were also found to

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have a greater correlation with familial early-onset prostate cancer, occurring in 3.1% of such patients, compared to 0.6% in late-onset, non-familial prostate cancer [92]. Karlsson et al. supported these findings by showing an odds ratio of 3.4 for the association of the *HOXB13* mutation G84E with early-onset familial prostate cancer, with a 33% risk of developing prostate cancer by age 80, compared to 12% of the general population [93]. Thus, HOXB13 mutations are a significant risk for prostate cancer diagnosis, particularly for early-onset familial cases.

ATM:

Mutations in the Ataxia Telangiectasia Mutated (*ATM*) gene are also associated with cases of hereditary prostate cancer. *ATM* encodes a protein kinase involved in the repair of DNA double-strand breaks, working downstream in the same repair pathway as BRCA1 and BRCA2. While *ATM* mutations are not as strongly associated with prostate cancer as *BRCA1* and *BRCA2*, they still carry an odds ratio of 1.8, with 1.6% of aggressive cases and 0.8% of nonaggressive cases having the mutation [94]. Karlsson et al. divided a series of *ATM* mutations into tier 1 (likely pathogenic) and tier 2 (possibly pathogenic). They established an odds ratio of 4.4 for tier 1, indicating an increased risk of early onset prostate cancer, and an odds ratio of 1.4 for tier 2 [84]. Zhang et al. discovered that *ATM* mutations have a role in immunosuppression by dysregulating immune control of tumor spread through LAG3 upregulation in CD4+ T-lymphocytes [95]. They reasoned that the mechanism was through decreased ATM, being unable to properly stimulate AMPK-alpha, which is responsible for supporting the binding of XBP1 and EGR2 transcription factors to the LAG3 promoter, involved in the inhibition of cytokine production and T-cell proliferation. Therefore, ATM mutations contribute to earlier onset and more aggressive prostate

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cancer through a multifaceted mechanism involving genomic instability and immune dysregulation.

CHEK2:

CHEK2 is a tumor suppressor gene that encodes a serine/threonine kinase, which supports p53-driven cell cycle arrest, a crucial component of DNA damage response, and is activated by ATM to initiate DNA damage response and repair [85]. Germline *CHEK2* variants are associated with increased risk of prostate cancer, especially aggressive disease. A comprehensive evaluation by Wu et al. reported *CHEK2* germline mutations in approximately 1.8-2.0% of unselected prostate cancer patients, with a higher prevalence (up to 2.28%) in cases with lethal prostate cancer compared to 1.65% in low-risk cases, indicating enrichment in severe disease [96]. Truncating *CHEK2* mutations have been shown to significantly increase the risk of lethal prostate cancer, with odds ratios of 3.4 in non-familial cases and 9.0 in familial cases. In contrast, missense mutations elevate metastatic risk with odds ratios of 1.61 (non-familial) and 1.97 (familial) [97]. A meta-analysis further confirmed the association between *CHEK2* mutations and increased prostate cancer risk, although no clear link was found with familial aggregation [98].

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Interplay of Genomic and Epigenomic Alterations on the Tumor Immune Microenvironment

Genomic and epigenomic alterations in prostate cancer cells can have broader pathobiological consequences, specifically shaping the immune tumor microenvironment. Accumulating evidence suggests that specific genomic alterations can affect the tumor immune landscape by promoting immune cell infiltration or immune exclusion. Alterations in BRCA1/2, ATM, and CHD1 can

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create genomic instability, increasing tumor mutational burden and neoantigen load, and activating the cGAS-STING pathway, promoting type I interferon signaling and immune cell infiltration [99-101]. However, a large majority of prostate cancers are microsatellite-stable (also referred to as non-MSI-H) but still exhibit a profoundly immunosuppressive tumor microenvironment, characterized by low cytotoxic T-cell infiltration, impaired antigen presentation, and a dominance of suppressive myeloid populations. In such cases, the immune microenvironment is shaped by recurrent genomic alterations (PTEN loss, MYC amplification, alterations in AR signaling, etc.). It is reported that PI3K/AKT activation due to PTEN loss leads to reduced T-cell trafficking and increased expression of immunosuppressive cytokines such as IL-6 and CCL2 [102]. MYC-driven transcriptional programs also impact the immune microenvironment by inducing immune exclusion through repression of genes required for effective T-cell recruitment and activation [103]. A role of MYC in PD-L1 expression on tumor cells and myeloid cells has also been reported, suggesting its role in T-cell exhaustion [104]. High AR activity is also shown to modulate interferon signaling and antigen presentation, and AR pathway alterations are suggested to underlie the general immune-exclusion phenotype of prostate tumors [105-107]. Conversely, AR inhibition is shown to transiently enhance immune infiltration [107]. Lineage plasticity associated with genomic alterations is also thought to modulate the immune system. NEPC tumors, in general, exhibit low MHC class 1 expression, minimal T-cell presentation, and high expression of immune-checkpoint-independent escape mechanisms. Similarly, microRNAs-mediated regulation of gene expression and their release into the extracellular space can profoundly influence the prostate immune landscape [108, 109]. miR-21, elevated in advanced and metastatic prostate cancer, targets PTEN and PDCD4 expression, promoting the expansion of myeloid-derived suppressor cells (MDSCs) and M2-polarized macrophages [110, 111]. miR-221/222, induced in CRPC,

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enhance immune evasion through modulation of cytokine signaling and AR bypass pathways [112]. Exosomal miR-141 and miR-375 are also associated with altered immune signaling, acting as systemic effectors beyond the primary tumor site [113].

Precision Therapy in Prostate Cancer.

Precision therapy in prostate cancer represents a transformative approach, potentially shifting the focus from traditional "one-size-fits-all" treatments to individualized strategies guided by each patient's unique molecular and clinical profile (**Table 2**). By leveraging advancements in genomics, imaging, and biomarker-driven diagnostics, clinicians may identify specific subgroups of patients who are most likely to benefit from targeted therapies, such as PARP inhibitors and immunotherapies, as well as qualify for biomarker driven clinical trials **(Fig. 4)**. The National Comprehensive Cancer Network and the American Society of Clinical Oncology recommend germline and somatic testing for all patients with metastatic prostate cancer [114]. Even in patients with positive germline testing for homologous recombination mutations, somatic testing remains useful for identifying other potential targets for treatment.

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Androgen Receptor Pathway Targeting:

Androgen receptor pathway inhibition (ARPI) remains a critical focus for castration-resistant prostate cancer. Enzalutamide and Abiraterone are commonly used second-generation drugs that target AR in patients after the failure of ADT [78]. Enzalutamide competitively inhibits the ligand-binding domain of the AR, whereas Abiraterone inhibits 17-α hydroxylase (CYP17A1) to deplete adrenal and intra-tumoral androgen synthesis [78]. These next-generation androgen inhibitors are often given alongside prednisone or, preferably, dexamethasone to reduce anterior pituitary ACTH

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production, downregulating adrenal synthesis of androgens [78]. While these agents are widely used and represent standard-of-care approaches, their integration into the broader framework of precision medicine is being increasingly recognized, largely due to growing emphasis on identifying molecular biomarkers that can inform and refine AR-targeted treatment strategies. Detection of AR splice variants, especially AR-V7, in circulating tumor cells has been associated with resistance to AR pathway inhibitors and poorer outcomes. Therefore, patients with AR-V7-positive tumors may derive limited benefit from AR-directed therapies [115-118]. Similarly, tumors exhibiting neuroendocrine differentiation or harboring specific genomic alterations (e.g., TP53, RB1, or PTEN loss) often display reduced AR signaling dependence and are associated with more aggressive disease and resistance to AR-targeted therapies [24, 119, 120]. Researchers recently developed a novel AR antagonist, TAS3681, which had activity against both AR-FL and AR-V subtypes. TAS3681 exhibited strong activity against AR-V7-positive tumors and against the enzalutamide-resistant F877L/T878A and H875Y/T878A mutations, as well as the doralutamide-resistant V716M and H875Y mutations [121]. In this context, AR-targeted therapy is slowly transitioning from a one-size-fits-all approach to a more nuanced, biomarker-guided application. However, identifying additional predictive biomarkers and resistance mechanisms is necessary to refine the use of AR pathway inhibitors and transition AR-targeted therapy from one-size-fits-all to true precision-based therapy.

PARP Inhibitors:

Knowledge of mutations in genes related to DNA repair has led to the use of PARP inhibitors for treating prostate cancer after relapse. PARP (poly ADP-ribose polymerase) is an enzyme that detects and repairs single-stranded DNA breaks [122]. Initial studies that brought PARP inhibitors

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to market focused on their use in later lines of therapy. The PROFOUND trial stratified CRPC patients, with prior progression on taxane therapy and an ARPI, into two cohorts: Cohort A, which harbors mutations in *BRCA1*, *BRCA2*, and *ATM*; and Cohort B, which includes mutations in 12 other homologous recombination repair genes [123]. Each cohort was randomized to receive a second ARPI (control) or a PARP inhibitor (Olaparib). Olaparib significantly improved progression-free survival (PFS), objective response rate, and time to pain progression compared to control in Cohort A, with a median PFS of 7.4 vs. 3.6 months (HR 0.34, P<0.001). PARP inhibitors were then investigated prior to the initiation of taxane therapy in patients with *BRCA1* or *BRCA2* mutations, as a subgroup analysis of the TRITON-3 study found an improvement in progression-free survival with a hazard ratio of 0.53 (95% CI, 0.37 to 0.77) when the PARP inhibitor rucaparib was compared to docetaxel therapy in patients with CRPC with prior progression on an ARPI [124].

Synergy between PARP inhibitors and oral ARPIs was explored in multiple trials, allowing PARP inhibitors to be evaluated in even earlier lines of therapy in both selected and unselected patient populations. The PROpel (NCT03732820) trial demonstrated a significant improvement (24.8 vs. 16.6 months) in radiographic progression-free survival with Olaparib plus Abiraterone, compared to placebo plus Abiraterone in first-line therapy metastatic CRPC therapy in an unselected population of patients (i.e., those with and without DNA damage repair pathway mutations) with no prior exposure to an ARPI [125]. TALAPRO-2 demonstrated a similar statistically significant improvement in radiographic progression-free survival as well as an improvement in overall survival in a similar patient population [126]. The BRCAAWAY trial showed that the Olaparib plus Abiraterone improved median progression-free survival in patients with *BRCA1* and *BRCA2*

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mutations (39 months) compared to Abiraterone (8.6 months) or Olaparib (14 months) alone in first-line therapy for metastatic CRPC with no prior exposure to an ARPI [127].

The advancement of PARP inhibitors into earlier lines of therapy has continued with the AMPLITUDE study, in which men with homologous recombination repair gene alterations were randomized to either (a) Niraparib plus Abiraterone acetate and prednisone or (b) placebo plus Abiraterone acetate and prednisone. In this study, in which the majority of men had *BRCA1* or *BRCA2* mutations, there was an improvement in radiographic progression-free survival (hazard ratio = 0.63; 95% confidence interval: 0.49-0.80; P = 0.0001) [128]. These data support the use of PARP inhibitors with oral ARPIs in earlier lines of therapy, especially in patients with homologous recombination mutations like *BRCA*, and represent a major step forward in precision therapy for prostate cancer.

PI3K/AKT/mTOR Pathway Inhibitor:

For patients with PTEN inactivation or other mutations in the PI3K/Akt/mTOR pathways impacting PTEN expression, targeted inhibitors have demonstrated clinical activity. One such agent, Capivasertib, inhibits Akt and has been studied in multiple randomized trials with advanced prostate cancer patients in both unselected and biomarker-driven populations [129-131]. In the Phase II ProCAID trial, capivasertib combined with docetaxel and prednisone did not improve the primary outcome of composite progression-free survival when compared with placebo plus docetaxel and prednisone in both the unselected patient population and those patients with either PTEN deficiency identified by immunohistochemistry (IHC) of tumor tissue and/or an eligible alteration in *PIK3CA*, *PTEN*, or *AKT1* by next-generation sequencing (NGS) of either tumor tissue

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or baseline plasma ctDNA [131]. In the Phase II RE-AKT trial of capivasertib combined with enzalutamide in men with metastatic CRPC with prior progression on both docetaxel and Abiraterone, capivasertib plus enzalutamide did not improve the primary outcome of a composite response rate endpoint in either the entire population or those patients with loss of PTEN expression by IHC [130]. Despite these negative results, a press release of a positive phase 3 trial in men with metastatic hormone-sensitive prostate cancer with PTEN deficiency by IHC has been released for the combination of capivasertib, abiraterone acetate, prednisone, and ADT compared to the control arm of placebo, abiraterone acetate, prednisone, and ADT [132]. A separate AKT inhibitor, ipatasertib, when combined with abiraterone acetate in men demonstrated an improvement in radiographic progression-free survival in men with metastatic castrate resistant prostate cancer with PTEN deficiency as assessed by IHC (hazard ratio [HR] 0·77 [95% CI 0·61-0·98]; p=0·034) but not in the entire study population which included men without PTEN deficiency (HR 0·84 [95% CI 0·71-0·99]; p=0·043; not significant at α=0·01)[133]. Despite an improvement in radiographic PFS in the PTEN-deficient patient population, capivasertib combined with Abiraterone did not demonstrate a statistically significant improvement in overall survival in the PTEN-deficient or the entire population [134].

EZH2 Inhibitors:

Inhibitors of the epigenomic regulator EZH2 represent an emerging class of targeted agents in prostate cancer, with relevance to molecular subtypes characterized by lineage plasticity, neuroendocrine differentiation, or resistance to ARPI. They have recently moved into phase 3 trials for patients with metastatic CRPC. Support for the phase 3 trial includes data from the phase ½ dose-escalation trial of mevrometostat with or without enzalutamide, with a median radiographic

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progression-free survival period of 17.0 months [135]. The phase 3 trials of mevrometostat are in unselected patients with advanced prostate cancer. While still under clinical investigation, EZH2-targeted therapy holds promise as a precision approach for genomically and epigenetically defined subsets of advanced prostate cancer.

Adoptive T-cell Therapies

Adoptive T-cell therapies represent a rapidly evolving class of immunotherapies that harness and redirect a patient's own immune cells to recognize and destroy tumor cells. These approaches include the use of genetically engineered T cells, such as chimeric antigen receptor (CAR) T cells and bispecific T-cell engagers (BiTEs), to overcome immune evasion mechanisms and enhance antitumor activity. Tarlatamab, a bispecific T-cell engager, is an approved therapy for small-cell lung cancer (SCLC) that directs cytotoxic T-cells to the DLL3 antigen commonly expressed in SCLC. Tarlatamab demonstrated an improvement in overall survival in patients with progressive SCLC when compared with second-line chemotherapy (hazard ratio for overall survival: 0.60; 95% CI, 0.47 to 0.77; P<0.001) [99]. NEPC may also express DLL3, making tarlatamab a potential therapy in this difficult-to-treat condition. In a phase Ib study of men with NEPC, patients with DLL3+ tumors treated with tarlatamab (vs. patients with DLL3-/DLL3 unknown tumors) achieved a higher overall response rate [22.2% (95% CI, 6.4-47.6) vs. 0% (95% CI, 0-15.4)] and radiographic progression-free survival rate at 6 months [27.7% (95% CI, 8.7-50.9) vs. 0%], suggesting DLL3 expression may predict response to tarlatamab in this patient population [136].

In an early-phase clinical trial (NCT03873805), patients with metastatic castration-resistant prostate cancer (mCRPC) received CAR T cells engineered to target the PSCA protein on tumor

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cells. On day 28 of CAR T cell infusion, 4 among 14 treated patients had prostate-specific antigen declines greater than 30% and showed radiographic improvements, indicating early signs of antitumor activity [137]. Another phase 1 clinical trial (NCT03089203) evaluated PSMA-targeted CAR T cells engineered to resist TGF-β–mediated immunosuppression in patients with mCRPC. Among 13 treated patients, four showed PSA reductions of ≥30%, and CAR T cells demonstrated *in vivo* expansion and tumor infiltration. Despite these signs of activity, resistance emerged through upregulation of inhibitory molecules in the tumor microenvironment. Stable disease was observed in 38.5% of patients at 3 months, with a median overall survival of approximately 16 months. These findings support advancing multipronged strategies to overcome tumor immunosuppression and further optimization of dosing and combination approaches to refine CAR T cell efficacy in prostate cancer [138].

Immune-Checkpoint Inhibitors:

Immune checkpoint inhibition is a promising precision therapy for prostate cancer, although evidence is currently lacking for the use of PD-L1 and CTLA-4 inhibitors as monotherapy or as combination therapy with chemotherapy or ARPIs in unselected patients. Pembrolizumab is approved for men with metastatic CRPC with microsatellite instability-high or mismatch repair-deficient tumors based on the KEYNOTE-164 study, which evaluated over 30 different types of tumors [139]. Retrospective data have demonstrated some nuances in the use of pembrolizumab in advanced prostate cancer. Lenis et al. showed that for patients with MSI-H/dMMR, treatment with PD-1 inhibitor pembrolizumab resulted in 45% of patients having a RECIST response, which measures the presence of target lesions, and 65% having a PSA50 response [82]. While no TMB-H/MSS patients presented with a RECIST response to immune checkpoint blockade, 50% had a

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significant PSA50 response. Supporting these results, MSI-H/dMMR had longer progression-free survival after treatment than TMB-H/MSS. Lenis et al. reasoned that this was due to an increased neoantigen burden and decreased genomic diversity in MSI-H dMMR compared to TMB-H/MMS. A similar study conducted by Graham et al. showed a PSA50 response in 53% of men and six-month progression-free survival in 64.1% of men with MSI-H/dMMR tumors after treatment with pembrolizumab [140]. It is worth noting that while both MSI-H/dMMR and TMB-high status are associated with increased neoantigen load and potential immunotherapy sensitivity, they represent distinct, sometimes overlapping, biomarker populations in prostate cancer [82]. Beyond MSI-H/dMMR, biallelic CDK12 inactivation also defines a prostate cancer subtype with genomic instability and high neoantigen load, promoting an immunogenic tumor microenvironment. Preclinical and clinical evidence suggest that CDK12-altered mCRPC may be more responsive to immune checkpoint blockade, supporting its potential as a predictive biomarker for immunotherapy [141]. Similarly, BRCA1/2-mutated prostate cancers may exhibit increased DNA damage and neoantigen presentation, providing a rationale for combination therapies pairing PARP inhibitors with immune checkpoint blockade [142, 143].

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Prospective evaluations of immune checkpoint inhibitors in patients with biomarkers suggesting an increased susceptibility to immunotherapy have included both single-agent immunotherapy and combination immunotherapy. In the Phase 2 Perseus1 trial, men with metastatic castrate resistant prostate cancer and either mismatch repair-deficient tumors or other prespecified biomarkers associated with a response to immune checkpoint inhibition had a composite response rate of 28% (95% CI, 12.1-49.1%) [144]. Combination treatment with PD-1 inhibitor nivolumab and the CTLA-4 inhibitor ipilimumab was investigated in the CheckMate 650 trial, and the results revealed

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response rates of 25% and 10% pre-chemotherapy and post-chemotherapy, respectively, in a population of patients with unselected metastatic castrate-resistant prostate cancer [145]. In the phase II NEPTUNES trial, in which > 85% of patients had previously been treated with chemotherapy, the combination of ipilumab 3 mg/kg and nivolumab 1 mg/kg was associated with a composite response rate of 40% (90% CI, 26%-55%) [146]. Given the potential for increased activity of combinations of PD-1 and CTLA-4 inhibitors compared to PD-1 inhibition alone, prospective randomized trials are needed to compare the efficacy of these two potential management strategies.

Tumor-restricted expression of B7-H3 (CD276) and its immunosuppressive role make it a promising target for immune checkpoint blockade. Preclinical studies demonstrate that B7-H3 inhibition can enhance T-cell–mediated antitumor immunity, while clinical strategies have explored both monoclonal antibodies and bispecific constructs to potentiate immune responses. Notably, B7-H3x4-1BB bispecific antibodies have been designed to localize 4-1BB co-stimulation to the tumor microenvironment, selectively activating tumor-infiltrating CD8+ T cells and enhancing cytotoxic activity while minimizing systemic immune-related toxicity [147]. Clinical studies suggest that combining B7-H3 blockade with other checkpoint inhibitors, such as PD-1 and CTLA-4, can enhance antitumor immunity. Notably, co-administration of enoblituzumab (anti-B7-H3) and pembrolizumab (anti-PD-1) demonstrated promising efficacy, indicating potential synergistic activation of T cells [148]. Preclinical studies further show that dual inhibition of B7-H3 with CTLA-4 or PD-L1 produces curative responses in murine models of castration-resistant prostate cancer, underscoring the value of combinatorial checkpoint strategies in molecularly defined tumor subtypes [149]. Collectively, these findings position B7-H3 as a

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versatile and promising immune checkpoint target, with both monotherapy and combination strategies demonstrating the potential to overcome tumor immune evasion, enhance T-cell-mediated cytotoxicity, and improve outcomes in molecularly defined, treatment-resistant prostate cancers.

Radioligand Therapies:

Prostatic-Specific Membrane Antigen (PSMA) is a common surface biomarker associated with prostate cancer, including CRPC, and can easily bind to radioisotopes. One of the most common radioisotopes used is lutetium-177 vipivotide tetraxetan (177-Lu-PSMA-617), which delivers beta radiation to PSMA-positive cells and surrounding tissues. The VISION trial investigated the efficacy and safety of 177-Lu-PSMA-617 for PSMA-positive castrate resistant prostate cancers following treatment with a taxane therapy and progression on an oral APRI [150]. Compared to a control arm of a second ARPI, 177-Lu-PSMA-617 was associated with an improvement in median progression-free survival of 8.7 months compared to 3.4 months, and an improvement in median overall survival of 15.3 months in the treatment group compared to 11.3 months in the control arm. The VISION study led to FDA approval of 177-Lu-PSMA-617 for PSMA-positive prostate cancer in 2023 [151], allowing men with this biomarker to receive the therapy. Analogous to PSMA-targeted approaches, other cell-surface proteins, such as STEAP1 and B7-H3, represent attractive targets for radioligand therapy and offer potential options for patients with PSMA-negative disease [152]. Thus, radioligand therapies represent an expanding next-generation precision strategy that leverages tumor-specific expression of cell-surface antigens, expanding the therapeutic toolkit beyond conventional systemic therapies.

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Antibody-drug Conjugates:

Antibody–drug conjugates (ADCs) have emerged as a promising precision medicine strategy, enabling selective delivery of potent cytotoxic agents to tumor cells while minimizing systemic toxicity. ADCs comprise three key components: i) a monoclonal antibody (mAb) binding a selective tumor antigen, ii) a cytotoxic payload, and iii) a linker that connects the mAb to the payload and controls its release. Upon binding to the target antigen on a cancer cell, the ADC is internalized, releasing the cytotoxic drug after intracellular processing, resulting in linker cleavage. Some ADCs also exhibit a bystander effect, in which the released payload diffuses into neighboring tumor cells, potentially overcoming intratumoral heterogeneity [153, 154]. Besides its use in radioligand therapy, PSMA has also been exploited for ADC-based therapies. A Phase I trial tested the efficacy of MEDI3726, a synthesized form of the anti-PSMA antibody site J591.87, and recorded strong antitumor activity. MEDI3726 binds the external region of PSMA to release a pyrrolobenzodiazepine dimer, which cross-links double-stranded DNA and triggers cell death. Another Phase I trial by Petrylak et al. used a PSMA antibody-drug conjugate and demonstrated its safety, with a maximum tolerated dose of 2.5 mg/kg [155]. A subsequent Phase II trial by the same group found a PSA50 response in 14% of all patients and 21% of patients who had never received chemotherapy [156].

Six-transmembrane epithelial antigen of the prostate-1 (STEAP-1) is a cell-surface protein predominantly expressed in prostate cancer, with minimal expression in normal tissues, making it a potential target for antibody–drug conjugates (ADCs). In a phase I/II study of patients with metastatic castration-resistant prostate cancer, PET/CT imaging with the radiolabeled anti–STEAP1 antibody 89Zr-DFO-MSTP2109A demonstrated high lesion detectability, identifying

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approximately 86% of histologically confirmed disease sites [157]. A subset of patients received the STEAP1-directed ADC DSTP3086S, which incorporates the microtubule inhibitor monomethyl auristatin E. Treatment was generally well tolerated but showed limited clinical activity, with modest responses and no correlation between tumor uptake and PSA changes, underscoring the need for improved ADC optimization and patient selection [158]. Trophoblast cell surface antigen-2 (TROP2) is also overexpressed in prostate cancer and associated with disease progression. Sacituzumab govitecan, a TROP2-directed ADC delivering SN-38, showed modest activity in a phase I/II basket trial, with an objective response rate of 9.1% and stable disease in 36% of patients with metastatic castration-resistant prostate cancer. Toxicities were mainly hematologic and manageable [159]. Several early-phase trials are ongoing to further evaluate TROP2-targeted ADCs, including next-generation agents such as datopotamab deruxtecan [160].

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The tumor-restricted expression of B7-H3 has also prompted the development of B7-H3-directed ADCs. DS-7300 is a B7-H3–directed ADC with a topoisomerase I inhibitor payload. A phase I/II study (NCT04145622) demonstrated manageable safety and early antitumor activity, including responses in heavily pretreated mCRPC patients. Common adverse events included nausea, infusion reactions, and vomiting, with higher-grade toxicities at the highest dose [161]. Vobra duo (MGC018) is another B7-H3-targeted ADC with a duocarmycin payload that has shown promising activity in metastatic castration-resistant prostate cancer. In the Phase 2 TAMARACK study, patients previously treated with AR-targeted and/or docetaxel therapies demonstrated meaningful PSA responses (37–44%) and objective responses (18–25%), with higher efficacy observed in taxane-naïve patients [162]. These results highlight the potential of B7-H3 ADCs to selectively

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deliver cytotoxic therapy to tumors, offering a novel treatment approach for advanced prostate cancer. These results support the continued clinical development of B7-H3 ADCs as a targeted cytotoxic therapy for advanced prostate cancer, particularly in earlier-line or high-risk patients, while emphasizing the need for careful monitoring and management of effusion-related toxicities.

Precision Therapy for Aggressive Variant Prostate Cancer with a Molecular Signature

In the somatic testing of advanced prostate cancer, the identification of deleterious changes in two of PTEN, TP53, and RB1 may be associated with an increased response to combination therapy with cabazitaxel and carboplatin. The combination of two such changes is classified as having an aggressive variant prostate cancer molecular signature (AVPC-MS). In a post-hoc analysis of a phase II trial comparing cabazitaxel to cabazitaxel plus carboplatin that enrolled men identified as having clinically aggressive variant prostate cancer, those men with the AVPC-MS criteria had a significant improvement in median progression-free survival 7.5 months (95% CI 4.4-9.6) versus 1·7 months (1·3 to NA)) and estimated median overall survival (20.2 months (13.3-37.2) versus 8·5 months (4·8 to NA), p=0·0002) [163]. An ongoing clinical trial, S2312, is evaluating carboplatin and cabazitaxel compared with cabazitaxel alone in both an unselected population of patients with metastatic CRPC and in men with AVPC-MS.

Acquired Resistance Mechanisms for Precision Therapy of Prostate Cancer

While precision therapy continues to advance prostate cancer management, secondary or acquired resistance can emerge to certain therapies due to pathway rewiring or transition into alternative phenotypic states. As discussed earlier, resistance to AR pathway inhibitors (ARPI) can emerge due to sustained activation of PI3K/AKT and other survival signaling pathways, many of which

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are associated with TP53/RB1/PTEN loss, MYC amplification, or neuroendocrine differentiation [24, 26, 119]. Resistance to PARP inhibitors is associated with restoration of DNA damage repair, often driven by reversion mutations in BRCA1/2 or ATM. These events reverse homologous recombination deficiency (HRD) and thus abrogate the synthetic lethality induced by PARP inhibitors [164, 165]. True ATM reversion is less common and ATM compensation occurs through partial restoration of ATM signaling and/or activation of compensatory DDR kinases such as ATR and DNA-PK [166, 167]. It is likely that the reversion mutations exist as low-frequency subclones prior to therapy and are positively selected under PARP inhibitor pressure; thus, serial genomic monitoring or switching to combination therapies targeting replication stress or cell cycle checkpoints may be required [168, 169]. The efficacy of 177-Lu-PSMA-617 heavily depends on homogeneous PSMA expression, and resistance often occurs when tumors have reduced or spatially heterogeneous PSMA expression [170, 171]. PSMA downregulation can occur via epigenetic mechanisms associated with altered chromatin accessibility and promoter methylation [172]. Lineage plasticity into AR-independent states, characterized by low or no PSMA expression, can also lead to secondary resistance [173, 174]. To overcome, serial PSMA PET imaging and the use of complementary tracers to identify PSMA-negative disease are required. Also, future development of strategies to restore PSMA expression via epigenetic modulation may help in combination therapy settings.

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Emerging Concepts and Challenges.

Precision oncology for prostate cancer is a rapidly developing field. Genomic testing following diagnosis has revolutionized treatment by stratifying patients based on biomarkers or molecular alterations that may be targeted by novel therapies or used to predict intrinsic resistance to specific

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treatments, thus saving unnecessary physical and economic burden. Further, in recent years, research has shifted from *in vitro* 2-D cultures to organoid models or 3-D cultures using patient-derived tissue samples that mimic the structure and function of the donor prostate [175]. These models allow for more patient-centered research and testing to advance precision treatment. Artificial intelligence (AI) is also emerging as a new method to assist in prostate cancer diagnosis and treatment planning. AI analyzes genomic trends across patients and treatment outcomes to improve personalized treatments based on genomic and somatic testing [176]. In parallel, multi-omics approaches, employing genomics, transcriptomics, and metabolomics to identify novel therapeutic targets and tailor more specific treatments, are also emerging rapidly [177]. Moreover, advances in next-generation genomic technologies, particularly single-cell sequencing and spatial genomics, are transforming precision therapy in prostate cancer by enabling high-resolution characterization of tumor heterogeneity, clonal evolution, and tumor-microenvironment interactions [178-180].

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Liquid biopsies, obtained through minimally invasive procedures from blood and other bodily fluids, are gaining traction to evaluate tumor biomarkers as opposed to more invasive traditional tumor biopsy techniques. For prostate cancer, liquid biopsies mainly test for circulating tumor DNA (ctDNA) and circulating tumor cells (CTC). These tests effectively detect actionable mutations, such as hereditary homologous recombination mutations, as well as other markers, such as PTEN [181]. Liquid biopsies also allow for the monitoring of possible neuroendocrine changes that would render androgen deprivation therapy futile [182]. Scanning ctDNA for neuroendocrine markers, such as synaptophysin and chromogranin, as well as for driver mutations in genes, such as RB1 and TP53, can provide a less invasive alternative for detecting neuroendocrine (NE)

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transdifferentiation [183]. Although no prospectively validated commercial panel for NE is widely implemented in clinical practice, an assay developed by Zhao et al. that uses a qPCR panel of NE markers and AR targets is being transitioned to CLIA-certified test status [184].

PSMA PET uses radiolabeled ligands targeting prostate-specific membrane antigen (PSMA), which is highly overexpressed in most prostate cancer cells. PET/MRI combines the molecular sensitivity of PSMA PET with the superior soft-tissue resolution of MRI. These techniques outperform conventional imaging (CT, bone scan) for the detection of lymph node and bone metastases and reduce the radiation exposure. Changes in PSMA uptake can reflect response to ADT and other AR-targeted therapies. These are superior to liquid biopsies, which do not localize metastases and are less sensitive in low-volume disease. The key limitations, however, are that PSMA could be downregulated following ADT or even lost in neuroendocrine tumors and has a limited ability to capture tumor heterogeneity at the molecular level. Indeed, there are challenges for all types of post-diagnosis testing of prostate cancer that need to be overcome. Somatic testing carries an array of issues, as somatic mutations can be extremely variable in cancer, with positivity for different mutations changing during tumor evolution [114]. This rapid evolution requires more frequent somatic testing, which in turn leads to increased costs from repeat biopsies. Increased costs can create barriers to effective treatment for some patients. Liquid biopsies, while effective and minimally invasive, have lower sensitivity than traditional biopsies [185]. Also, although less invasive, liquid biopsy costs are similar to those of conventional biopsies [114]. These barriers call on researchers to develop minimally invasive, cost-effective screening tools with sensitivity equivalent to that of traditional core biopsies.

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Conclusion

While we have identified several major genomic alterations in prostate cancer, not all have been established as effective therapeutic targets. This is further complicated by the existence of large intra-/inter-tumor and host heterogeneity, making the clinical management more challenging. Despite this, the progress has been remarkable, raising hope for continued advancements in molecular information-guided prostate cancer stratification and evolving strategies for treatments. For example, *BRCA1*/*2*and other homologous recombination gene mutations qualify patients for treatment with PARP inhibitors, which further inhibit additional DNA repair mechanisms, leading to improved therapeutic outcomes. *PTEN* mutations are also critical, as they activate the PI3K/AKT/mTOR growth pathway, potentially making patients vulnerable to treatments with PI3K/AKT inhibitors. Mutations causing microsatellite instability and mismatch repair deficiency, as well as the presence of a high tumor mutational burden, create therapeutic opportunities for treatment with immune checkpoint inhibitors, such as PD-1 inhibitors, due to increased neoantigen production. These biomarkers are among many that may be effectively targeted by novel personalized treatments. Continued research is necessary to develop and refine effective treatments for biomarker-driven therapy to expand the population of patients who may benefit from precision medicine in prostate cancer.

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LIST OF ABBREVIATIONS

ADT: Androgen Deprivation Therapy

AKT: Protein Kinase B (also known as PKB)

AMPK: AMP-Activated Protein Kinase

AR: Androgen Receptor

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AR-FL: Androgen Receptor Full-Length

ARPI: Androgen Receptor Pathway Inhibitor

AR-V7: Androgen Receptor Splice Variant 7

ASCL1: Achaete-Scute Family BHLH Transcription Factor 1

ATM: Ataxia Telangiectasia Mutated

AURKA: Aurora Kinase A

BHLH: Basic Helix-Loop-Helix

BRCA1/2: Breast Cancer Gene 1 and 2

CA: Caucasian American

CHEK2: Checkpoint Kinase 2

CHGA: Chromogranin A

CRPC: Castration-Resistant Prostate Cancer

CTLA-4: Cytotoxic T-Lymphocyte-Associated Protein 4

DDR: DNA Damage Repair

dMMR: Deficient Mismatch Repair

EMT: Epithelial-Mesenchymal Transition

ERG: ETS-Related Gene

ERRα: Estrogen-Related Receptor Alpha

ETS: Erythroblast Transformation-Specific Family of Transcription Factors

EZH2: Enhancer of Zeste Homolog 2

FOXA1: Forkhead Box A1

FOXA2: Forkhead Box A2

GSTP1: Glutathione S-Transferase Pi 1

HGPIN: High-Grade Prostatic Intraepithelial Neoplasia

HR: Hazard Ratio

IHC: Immunohistochemistry

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IFN: Interferon

LAG3: Lymphocyte Activation Gene 3

MHC: Major Histocompatibility Complex

miRNA /miR: MicroRNA

MLH1: MutL Homolog 1

MMAC1: Mutated in Multiple Advanced Cancers 1 (alternative name for PTEN)

MMP9: Matrix Metalloproteinase 9

mTOR: Mechanistic Target of Rapamycin

MSI-H: Microsatellite Instability-High

MSH2: MutS Homolog 2

MYCN: N-Myc Proto-Oncogene

NE: Neuroendocrine

NEPC: Neuroendocrine Prostate Cancer

NEUROD1: Neuronal Differentiation 1

NGS: Next-Generation Sequencing

NKX3.1: NK3 Homeobox 1

NLRP3: NOD-, LRR-, and Pyrin Domain-Containing Protein 3 (context-dependent, not explicitly used but sometimes related)

OPHN1: Oligophrenin 1

OXPHOS: Oxidative Phosphorylation (if mentioned later in text)

PARP: Poly (ADP-Ribose) Polymerase

PDGF-β: Platelet-Derived Growth Factor Beta

PD-1: Programmed Cell Death Protein 1

PD-L1: Programmed Death-Ligand 1

PFS: Progression-Free Survival

PI3K: Phosphoinositide 3-Kinase

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PLXNA2: Plexin A2

PMS2: Postmeiotic Segregation Increased 2 (related to dMMR/MSI-H context)

PSA: Prostate-Specific Antigen

PSA50: ≥50% Decline in Prostate-Specific Antigen

PTEN: Phosphatase and Tensin Homolog

RB1: Retinoblastoma 1

SPOP: Speckle-Type POZ Protein

SEMA3C: Semaphorin 3C

SOX2: SRY-Box Transcription Factor 2

SYP: Synaptophysin

TMB: Tumor Mutational Burden

WNT: Wingless/Integrated Signaling Pathway

XBP1: X-Box Binding Protein 1

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Author contributions

Ajay P. Singh and John C. Henegan conceived and designed the manuscript. John A. Ligon conducted the literature review, performed data analysis, and prepared the first draft. Shashi Anand, Seema Singh, Jawed Siddiqui, John C. Henegan, and Ajay P. Singh reviewed the draft and suggested revisions, including the addition of relevant literature. Shashi Anand and Seema Singh contributed to the graphic design. John A. Ligon and Shashi Anand prepared the final revised version of the manuscript. All authors reviewed and approved the final manuscript for submission.

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Data availability

All publicly available information was used and analyzed to prepare this review article.

Ethics approval

Not Applicable.

Competing interest

The authors declare that there are no competing interests.

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Figure legends

Figure 1: Genetic and epigenetic alterations across the disease states during prostate cancer progression. Prostate cancer likely progresses from high-grade prostatic intraepithelial neoplasia to prostate adenocarcinoma and invasive metastatic disease. While androgen-deprivation therapy is initially effective for advanced cases, many patients develop aggressive, castration-resistant prostate cancer (CRPC). This progression is driven by diverse genetic alterations (gene deletion, amplification, mutation/fusion), epigenetic modifications, and alternative splice variants linked to disease stage and treatment response. Created in BioRender. Anand, S. (2026) https://BioRender.com/d0vcwot.

Figure 2: Mechanism of neuroendocrine differentiation and lineage plasticity in prostate cancer. Neuroendocrine differentiation in prostate cancer involves the transformation of adenocarcinoma cells into neuroendocrine-like cells, often marked by androgen receptor (AR) loss and resistance to androgen-deprivation therapy. This process is driven by the loss of tumor suppressors (RB1, TP53), upregulation of transcription factors (ASCL1, NEUROD1, SOX2, FOXA2, BRN2), and epigenetic remodeling mediated by EZH2. Amplification of MYCN and AURKA further promotes neuroendocrine features and resistance to therapy. Created in BioRender. Anand, S. (2026) https://BioRender.com/qi9uuzh.

Figure 3: Key germline mutations and their role in hereditary prostate cancer. Germline mutations, key predictors of hereditary cancer risk, are inherited genetic changes present in the egg or sperm that become part of the DNA in every cell of the offspring. Both sperm and egg can equally contribute to germline mutations that lead to hereditary prostate cancer. For example, a

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mother or father carrying a mutation in *BRCA1*, *BRCA2*, *HOXB13*, *ATM*, or *CHEK2* can pass it on to their son, who will have a higher risk of developing prostate cancer. Many of these mutations are involved in DNA repair pathways, making them attractive targets for therapeutic intervention. Most hereditary prostate cancer mutations are inherited in an autosomal dominant manner, meaning that just one copy of the mutated gene (from either parent) is sufficient to increase risk. Created in BioRender. Anand, S. (2026) https://BioRender.com/k3kecd7.

Figure 4: Overview of various precision therapeutic approaches in prostate cancer. Precision therapy in prostate cancer involves personalized treatment strategies guided by each patient's distinct molecular and clinical profile. These strategies are informed by the unique biological signatures driving cancer progression and are made possible through advances in genomics, imaging, and biomarker-based diagnostics. Such approaches enable the identification of patient subgroups most likely to benefit from targeted therapies, including PARP inhibitors, PI3K/AKT/mTOR pathway inhibitors, radioligand and PSMA-targeted therapies, adoptive T-cell-based therapies, and immune checkpoint inhibitors. *-denotes drugs approved by the FDA for precision therapy in prostate cancer. Created in BioRender. Anand, S. (2026) https://BioRender.com/u3mir8z.

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Table 1: Common genetic alterations in Prostate Cancer

Category	Gene(s)	Type of Alteration	Frequency / Prevalence	Functional / Clinical Significance	Ref.
Gene Fusions	*TMPRSS2 -ERG*	Fusion of androgen-responsive TMPRSS2 with ETS transcription factor ERG	~50% of prostate cancers	Promotes tumor aggressiveness, vascular density, maintains AR signaling; promotes metastasis & castration resistance	[5-9]
	TMPRSS 2-ETV1 / ETV4 / ETV5	Gene fusions with ETS family members	Less common; mutually exclusive with TMPRSS2-ERG	Alternative ETS-driven oncogenic pathways; contribute to tumorigenesis in a subset of cases	[10-12]
Gene Mutations	*SPOP*	Point mutation/protein loss	Mutated in ~10-15%; loss in 42.9% cases, up to 60% in Grade 5	Loss leads to accumulation of AR and c-MYC; activates PI3K/mTOR; associated with PTEN loss, high-grade tumors, and perineural invasion	[13-16]
	FOXA1	Point mutation	~35% of CRPC	Dysregulates chromatin remodeling; induces SEMA3C (growth factor); associated with SPOP; contributes to biochemical recurrence & WNT dysregulation	[17-21]
	TP53	Loss-of-function mutations	Common in advanced disease	Associated with poor survival (↓ 1-, 3-, 5-year OS by 13%, 20%, 16%); causes resistance to antiandrogen therapy (via SOX2, EZH2); promotes metastasis	[22, 24]
	RB1	Mutation or loss	Frequent in advanced & NEPC	Loss promotes metastasis and phenotype evolution; contributes to therapy resistance	[23, 24]

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Gene Deletions	*PTEN* (10q23.3)	Deletion/loss of heterozygosity/point mutation	Common in late-stage & metastatic prostate carcinoma	Negative regulator of PI3K/Akt/mTOR; loss promotes AR blockade resistance, immune suppression; high heterogeneity among metastatic sites	[25-28]
	NKX3.1 (8p21.2)	Deletion/downregulation	Common in prostate cancer	Loss impairs differentiation & DNA repair; promotes TMPRSS2-ERG fusion; interacts with AR/AKT pathway; contributes to castration resistance	[29-32]
	CHD1	Focal deletion (5q21)	Significant subset; ETS-negative tumors	Alters chromatin accessibility and AR-driven transcription; impairs homologous recombination; induces BRCA-like phenotype and PARP inhibitor sensitivity	[33-39]
Gene Amplifications	*MYC*	Amplification/overexpression	~29% of PCa; higher (66%) in Gleason ≥8; more frequent in African descent	Drives aggressiveness, castration resistance, and immune evasion; disrupts AR signaling; predictor of metastasis	[40-44]
	AR	Amplification	Frequent in CRPC	Enables AR signaling under low androgen; causes resistance to enzalutamide/abiraterone; promotes progression	[45-47]
	MYB	Amplification/overexpression	26% overall; 53% in hormone-refractory disease	Enhances AR activity in low androgen; drives aggressiveness & early recurrence; more prevalent in Black/African American patients	[48-53]
	PYGO2	Amplification	Advanced disease	Wnt/β-catenin chromatin effector; promotes lineage plasticity, immune	[54-59]

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				evasion, and epigenetic reprogramming	
Epigenomic Alterations	*EZH2*	Overexpression (histone methyltransferase activity)	Positive in 56.6% of PCa	Correlates with high Gleason grade, early PSA recurrence, TMPRSS2–ERG fusion; promotes chromatin silencing	[60]
	GSTP1	Promoter hypermethylation /gene silencing	>90% of PCa	Loss causes oxidative DNA damage accumulation; marker for oxidative-stress-based therapies	[61-63]
	miRNAs (e.g., miR-150)	Epigenetic/posttranscriptional regulation	Dysregulated in PCa	High androgen induces miR-150 → represses MYB; regulates androgen biphasic response; miRNAs serve as biomarkers via extracellular vesicles	[65, 66]
	Multiple miRNAs (e.g., miR-1, miR-21, miR-29b, miR-200)	Dysregulated miRNA expression	Stage-specific	Regulate EMT, angiogenesis, anoikis resistance, immune interactions; involved in progression and therapeutic response; potential biomarkers	[64, 66-70]

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Table 2: Precision therapies in Prostate cancer

Therapeutic Class	Drug(s)	Molecular Target / Mechanism	Predictive Biomarker(s) / Eligible Population	Key Clinical Trials / Ref.
Androgen Receptor Pathway Inhibitors (ARPI)	Enzalutamide	Competitive inhibition of AR ligand-binding domain	AR-dependent CRPC; reduced benefit in AR-V7-positive tumors	[78, 115-118]
	Abiraterone (+prednisone/ dexamethasone)	CYP17A1 inhibition → depletion of adrenal/intratumoral androgens	AR-driven disease; less effective in AR-independent tumors	[78]
	TAS3681	Novel AR antagonist targeting AR-FL and AR splice variants	AR-V7-positive tumors; AR mutations (F877L, T878A, H875Y, V716M)	[121]
PARP Inhibitors (monotherapy)	Olaparib	PARP inhibition → synthetic lethality in HRR-deficient tumors	BRCA1/2, ATM mutations; mCRPC post-ARPI and taxane	PROFOUND [123]
	Rucaparib	PARP inhibition	BRCA1/2 mutated mCRPC post-ARPI	TRITON-3 [124]
PARP Inhibitors + ARPI (Combination)	Olaparib + Abiraterone	Dual AR suppression + DNA repair inhibition	Unselected mCRPC; enhanced benefit in HRR-mutated tumors	PROpel [125]
	Talazoparib + Enzalutamide	PARP + AR blockade	First-line mCRPC	TALAPRO-2 [126]
	Olaparib + Abiraterone	Same as above	BRCA1/2-mutated mCRPC	BRCAaway [127]
	Niraparib + Abiraterone	Same as above	HRR-mutated mCRPC (mostly BRCA1/2)	AMPLITUDE [128]
PI3K/AKT/ mTOR Pathway Inhibitors	Capivasertib, Ipatasertib	AKT inhibition	PTEN deficiency (by IHC), PIK3CA, AKT1 alterations	ProCAID [131], RE-AKT [130]
	Ipatasertib + Abiraterone	AKT inhibition	PTEN-deficient mCRPC	IPATential150 [133]
EZH2 Inhibitors	Mevrometostat with or	Epigenetic regulator; histone	Neuroendocrine differentiation, lineage	Phase 1/2 and Phase 3 ongoing [135]

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	without enzalutamide	methyltransferase, EZH2 inhibition	plasticity, ARPI-resistant subtypes	
Adoptive T-cell / Bispecific Therapies	Tarlatamab (BiTE) PSCA CAR-T cells PSMA CART cells	DLL3-directed T-cell engager CAR T-cell cytotoxicity CAR T resistant to TGF-β suppression	DLL3+ NEPC PSCA-expressing mCRPC PSMA+ mCRPC	(Phase Ib) [136] PSCA-CAR-T (NCT03873805) [137] PSMA-CAR-T (NCT03089203) [138]
Immune Checkpoint Inhibitors (ICI)	Pembrolizumab Nivolumab + Ipilimumab	PD-1 blockade PD-L1, CTLA-4 inhibition	MSI-H, dMMR mCRPC; selected TMB-H (tumor mutational burden–high) Selected biomarker-positive or unselected mCRPC	KEYNOTE-164 [139], PERSEUS1 [144] CheckMate 650 [145], NEPTUNES, [146]
Radioligand / Targeted Radiotherapies	177Lu-PSMA-617 (Pluvicto)	Beta-emitting PSMA-targeted radioligand therapy	PSMA-positive (by PET imaging) mCRPC	VISION trial [150, 151]
PSMA-Targeted Antibody-Drug Conjugates (ADC)	MEDI3726, PSMA-ADC STEAP1 ADC (DSTP3086S) Sacituzumab govitecan B7-H3 ADCs (DS-7300, MGC018)	PSMA-targeted antibody linked to cytotoxic payload STEAP1-targeted MMAE delivery TROP2-targeted SN-38 delivery B7-H3–targeted cytotoxic delivery	PSMA-positive mCRPC STEAP1+ mCRPC TROP2+ mCRPC B7-H3-expressing mCRPC	Phase I/II trials [155, 156] [158] [159] [161, 162]
Molecularly defined Chemotherapy	Cabazitaxel + Carboplatin	Cytotoxic synergy; platinum sensitivity in genomically defined subsets	AVPC-MS (loss of ≥2 of TP53, RB1, PTEN)	Phase II post-hoc analysis; S2312 trial [163]

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Figure 1

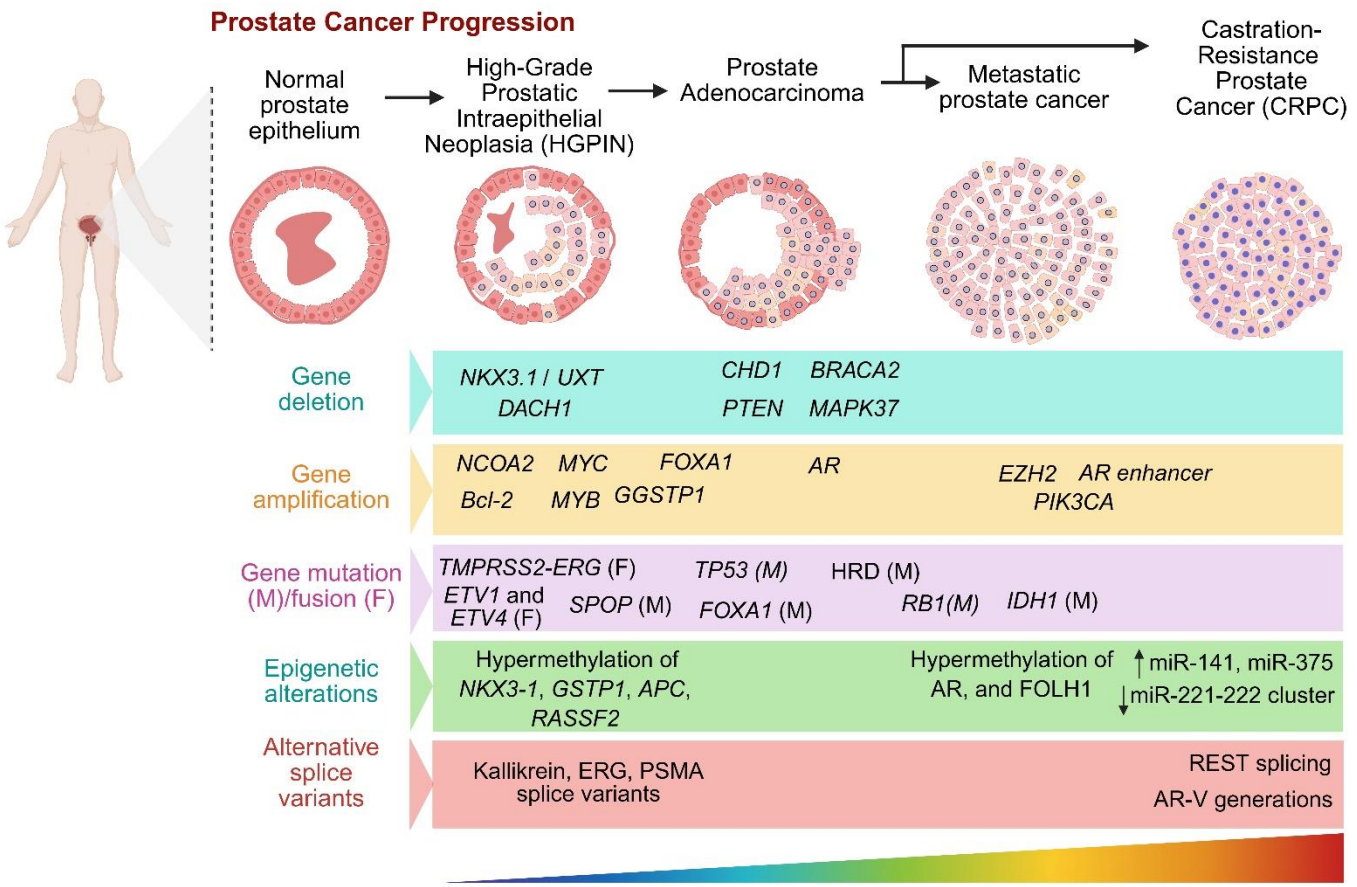

Prostate Cancer Progression
Normal prostate epithelium
High-Grade Prostatic Intraepithelial Neoplasia (HGPIN)
Prostate Adenocarcinoma
Metastatic prostate cancer
Castration-Resistance Prostate Cancer (CRPC)
Gene deletion
NKX3.1 / UXT
DACH1
CHD1
PTEN
BRACA2
MAPK37
Gene amplification
NCOA2
Bcl-2
MYC
MYB
FOXA1
GGSTP1
AR
EZH2
AR enhancer
PIK3CA
Gene mutation (M)/fusion (F)
TMPRSS2-ERG (F)
ETV1 and ETV4 (F)
SPOP (M)
TP53 (M)
FOXA1 (M)
HRD (M)
RB1(M)
IDH1 (M)
Epigenetic alterations
Hypermethylation of NKX3-1, GSTP1, APC, RASSF2
Hypermethylation of AR, and FOLH1
miR-141, miR-375
miR-221-222 cluster
Alternative splice variants
Kallikrein, ERG, PSMA splice variants
REST splicing
AR-V generations

Figure 2

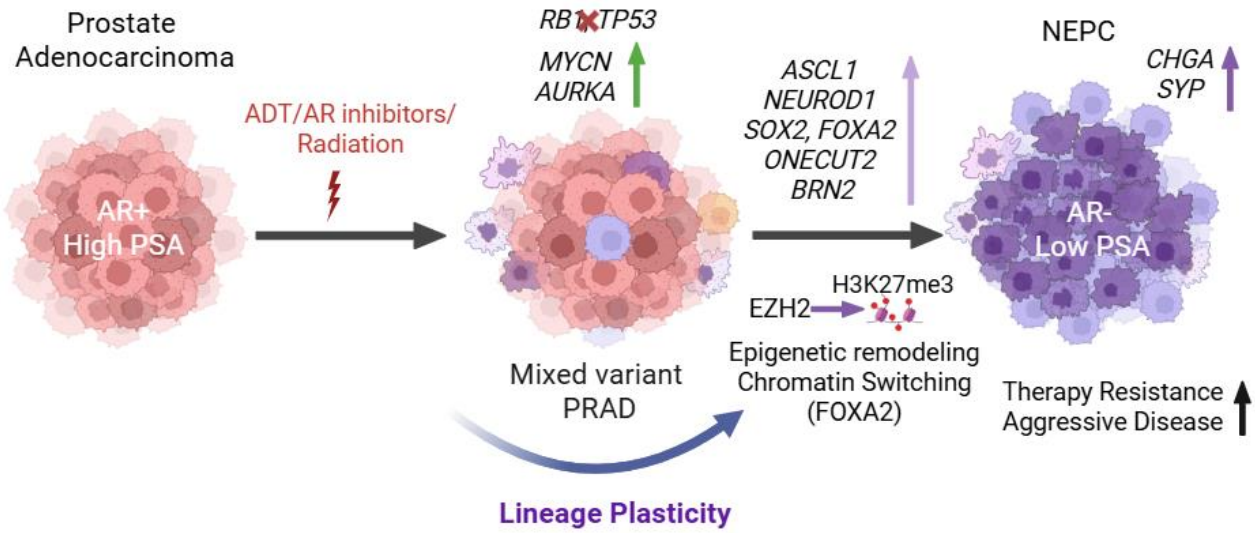
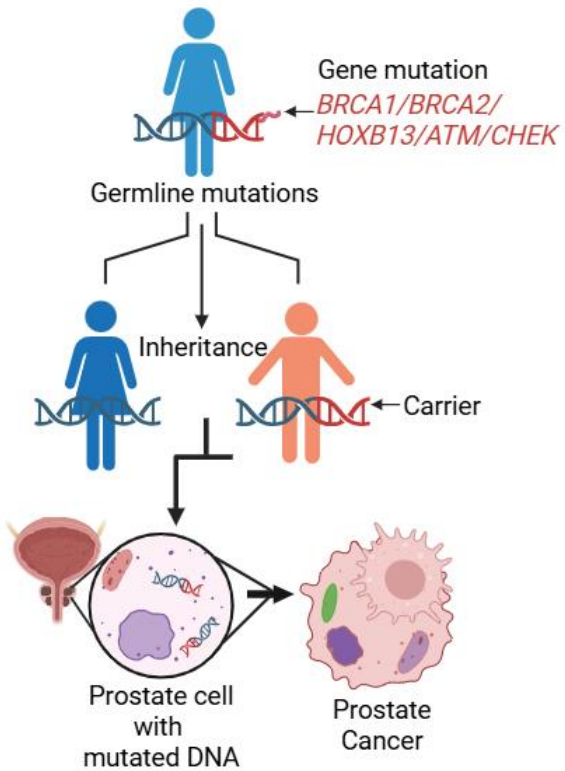
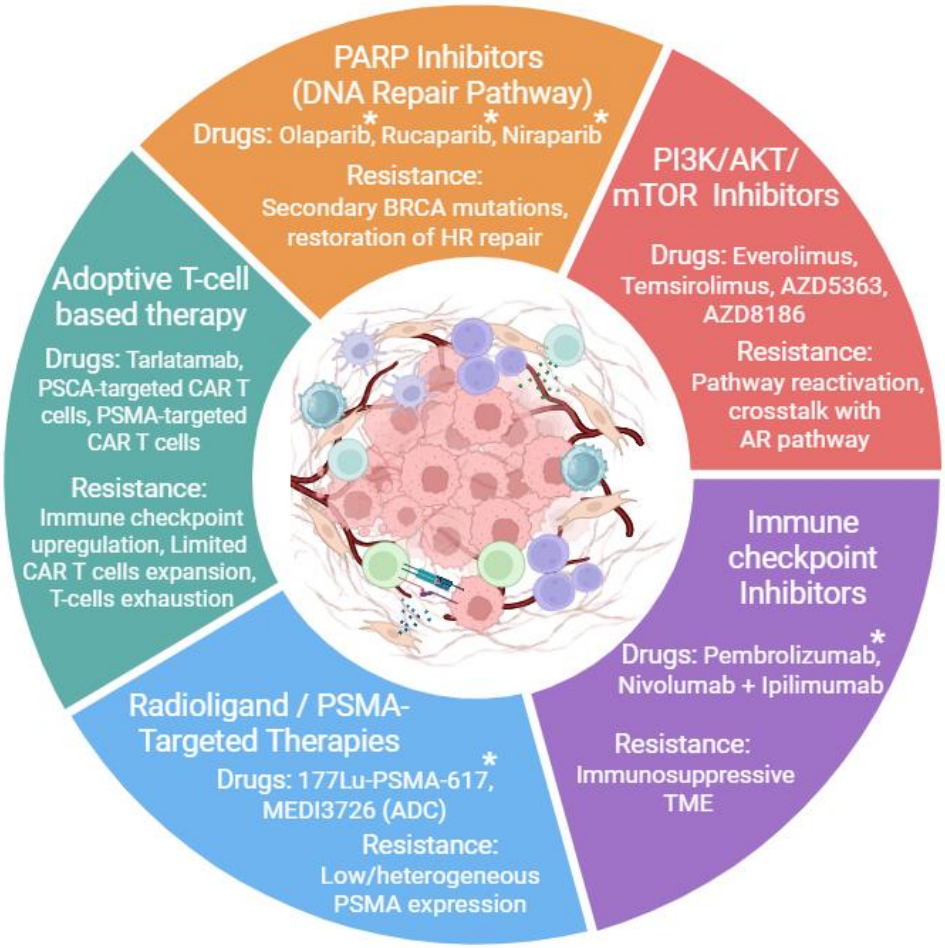

Prostate
Adenocarcinoma
AR+
High PSA
ADT/AR inhibitors/
Radiation
RB
TP53
MYCN
AURKA
Mixed variant
PRAD
ASCL1
NEUROD1
SOX2, FOXA2
ONECUT2
BRN2
EZH2
H3K27me3
Epigenetic remodeling
Chromatin Switching
(FOXA2)
NEPC
CHGA
SYP
AR-
Low PSA
Therapy Resistance
Aggressive Disease
Lineage Plasticity

Figure 3


Gene mutation
BRCA1/BRCA2/
HOXB13/ATM/CHEK
Germline mutations
Inheritance
Carrier
Prostate cell
with
mutated DNA
Prostate
Cancer

AK

Figure 4


PARP Inhibitors
(DNA Repair Pathway)
Drugs: Olaparib*, Rucaparib*, Niraparib*
Resistance:
Secondary BRCA mutations, restoration of HR repair
PI3K/AKT/
mTOR Inhibitors
Drugs: Everolimus, Temsirolimus, AZD5363, AZD8186
Resistance:
Pathway reactivation, crosstalk with AR pathway
Immune
checkpoint
Inhibitors
Drugs: Pembrolizumab*, Nivolumab + Ipilimumab
Resistance:
Immunosuppressive TME
Radioligand / PSMA-
Targeted Therapies
Drugs: 177Lu-PSMA-617*, MEDI3726 (ADC)
Resistance:
Low/heterogeneous PSMA expression
Adoptive T-cell
based therapy
Drugs: Tarlatamab, PSCA-targeted CAR T cells, PSMA-targeted CAR T cells
Resistance:
Immune checkpoint upregulation, Limited CAR T cells expansion, T-cells exhaustion